Proc. 12th International Conference on Electromagnetic Wave Interaction with Water and Moist Substances (ISEMA), Lublin, Poland, 4-7 June, Ed: Skierucha, 106-109.

Electromagnetic Characterization of Coarse-grained Soils with a One Port Large Coaxial Cell

Thierry Bore, Habibullah Bhuyan, Tilman Bittner and Alexander Scheuermann
School of Civil Engineering
University of Queensland
St Lucia, Australia
t.bore@uq.edu.au

Norman Wagner
Institute of Material Research and Testing
Bauhaus-University Weimar
Weimar, Germany

***Abstract*—This paper reports the design, the calibration and the application of a one port large coaxial cell. The objective of the device is to characterize the electromagnetic properties of compacted and partly saturated coarse grained soil under controlled boundary conditions. The dielectric spectra were obtained by means of fitting the measured scattering function using a Transvers electromagnetic mode (TEM) propagation model considering the frequency dependent complex permittivity. The method was applied on a construction site material with aggregate up to 15 mm. 6 samples were characterized, the results showing a good agreement in terms of Root Mean square error.**

***Keywords*—coaxial cell, coarse grained soils, relaxation model**

I. Introduction

Precise knowledge of the frequency dependent electromagnetic properties of coarse grained materials is mandatory for the successful application of high frequency electromagnetic measurement techniques for near and subsurface monitoring. The classic example is the application of Ground Penetrating Radar (GPR), which requires the perfect knowledge of electromagnetic properties to localize disturbances in space rather than in time ([1], [2]).

The electromagnetic characterization of coarse grained materials has to face an important challenge: the respect of the representative elementary volume (REV) criteria [3]. Previous studies have shown that material sample dimensions need to be at least three times greater the maximum dimensions of the major aggregate. In most applications, transmission or reflection line systems in transverse electromagnetic mode (TEM) are currently used to assess the relative complex permittivity. In these conditions, a compromise between the upper frequency limit of use and the maximal size of the het-heterogeneities has to be made. Otherwise, it is worth to note that most of the studies of the dielectric properties performed on coarse grained soil were performed in time domain ([4], [5]). As far as the authors know, only one study has been focused on the measurement of the frequency dependent dielectric properties of coarse grained soil [6]. Nevertheless, the authors were measuring in time domain before applying inverse discrete fast Fourier transform (DFFT). Previous studies have shown that measuring in frequency domain with Vector Network Analyser provides more accurate results [7]. In this framework, the development of experimental device that could provide frequency dependent dielectric properties of coarse grained materials from measured scattering functions in frequency domain represents an important goal.

The objective of this paper is to introduce a new methodology to obtain the broadband complex relative permittivity of compacted and partly saturated coarse grained soils. The first part will focus on the design of the coaxial cell. The method and the model used for the calibration will be briefly presented. In the second part, an example of application on a compacted and partly saturated coarse grained soil will be presented. The results provide a robust estimation in terms of root mean square error (RMSE) and demonstrate the capability of such a large coaxial cell for dielectric soil characterization.

II. A Large Coaxial Cell

A. Design

The materials targeted for our application can present a large particle size of up to 25 mm. In this configuration, the design of a large coaxial cell is required. The second main design constraints consist in having the possibility to compact the soil directly in the measurement device. In order to comply with existing compaction tests from geotechnical engineering, at least three layers of compacted material need to be placed in the cell with the height of the layers being approx. 3 times the maximum particle size. To allow compaction of the soil and to simplify the overall design, a one port measurement cell was selected. The large coaxial cell was made with three different coaxial line sections: a feeding section (which is mainly a commercial connector), a sealing system and the probe section where the material to be characterized is inserted (see Fig. 1 for illustration). The probe section presents the following transversal dimensions: diameter of inner conductor 66 mm, diameter of outer conductor 151.9 mm; whereas the length is 207.5 mm. Measurements are performed over the 1 MHz – 860 MHz frequency range. The upper frequency was set according to the cut off frequency (frequency limit of the line is defined as the frequency of the first higher order mode, i.e. when the propagation of electromagnetic wave within the cell is no more pure Transverse Electromagnetic mode (TEM)).

978-1-5386-4593-2/18/$31.00 ©2018 IEEE

B. Calibration and validation

Dielectric spectroscopy in such a non-uniform transmission line requires a wave propagation model that accounts for multiple reflections. Assuming a TEM propagation and non-magnetic materials, impedance transformation in combination with layer peeling algorithm was used to develop such a model [6]. Instead of using the classic parametrization with per-unit length parameters r, l, g and c to compute the propagation factor γ, the characteristic impedance Z_C and the resistance correction loss factor A; each section of the coaxial cell was parametrized with physical length d, geometric impedance Z_0, the material properties $\varepsilon_r(\omega)$ (we do not consider magnetic material, thus $\mu_r=1$) and the resistance factor α_R. This parametrization is interesting because it separates effects of geometric characteristic, material properties and electrical resistance instead of coupled line parameters. Complete details on this parametrization can be found elsewhere [8].

Finally, the forward model compute the scattering function $S_{11}(\omega)$ as a function of frequency and the 12 parameters of the line. The calibration step consists of determining the unknowns of the model using measurements in perfectly known materials. Measurements in air and deionized water were used. For air, a value of 1 was used for the complex permittivity, for deionized water the theoretical relative complex permittivity was computed as function of temperature thanks to the model proposed in [9]. Please note that in this configurations we aim at determining 11 parameters since the relative complex permittivity of calibration materials are known. This was achieved by minimizing the cost function between the measured and modeled scattering function. The quality of this step was validated with a good Root Mean Square Error (RMSE) and the obtained parameters were physically meaningful [10].

Fig. 1. Picture of the coaxial cell.

The calibration results were used as an input in the following to compute the dielectric spectrum of standard liquid to validate our approach. 2 liquids were used: salt water (conductivity σ_{DC} = $65.2\ mS.m^{-1}$) and ethanol-water mixture (fraction of ethanol x_e $=0.34$). The frequency dependent relative complex permittivity obtained were in good agreement with data from literature.

III. APPLICATION TO PARTLY SATURATED COMPACTED SOIL

A. Materials

A sample from the University of Queensland lakes construction site, synthetic football field and car park, was taken for testing. Particle size distribution (PSD) demonstrates that 5% material passes through the 0.075 mm sieve and 95% material passes through the 9.5 mm sieve. In overall, the site material was little cohesive in nature with maximum particle size 15 mm. The soil sample is prepared with oven drying procedure for the laboratory investigation. The proctor compaction test shows that the material has the optimum moisture content (OMC) 12% and maximum dry density 1290 kg/m3. According to the OMC, moisture content varies in the sample during electromagnetic characterization as shown in Table1 although the density is remained constant (80 % of MDD) because of the limitation of soil densification in the cell.

TABLE I. STATE PARAMETERS OF THE SAMPLES UNDER INVESTIGATIONS

Sample	*Gravimetric water content w [%]*	*Dry density ρ_D [kg.m^{-3}]*
S1 (6WC80D)	6.37	1609.46
S2 (8WC80D)	8.10	1583.72
S3(10WC80D)	9.99	1556.50
S4 (12WC80D)	11.32	1537.91
S5 (14WC80D)	12.99	1515.18
S6 (16WC80D)	16.30	1472.05

B. Results

In order to compute the relative complex permittivity from the measured scattering function, a model of frequency dependence of the permittivity was used. Generalized Dielectric Relaxation (GDR) model with 2 relaxations processes was used (1). Generally for high frequency measurement on porous media, three relaxation process are expected: free water relaxation, and 2 relaxation mechanisms which are related to interactions between the aqueous phases and solid particles (absorbed water, Maxwell Wagner effect and counter ion relaxation). Since free water relaxation occurs at frequency around 20 GHz, we used 2 models which include 2 Cole Cole terms to account for the interfaces process:

$$\varepsilon_r(\omega) = \varepsilon_\infty + \frac{\Delta\varepsilon_{IF}}{1+(j\omega\tau_{IF})^{\beta_{IF}}} + \frac{\Delta\varepsilon_{LF}}{1+(j\omega\tau_{LF})^{\beta_{LF}}} - j\frac{\sigma_{DC}}{\omega\varepsilon_0} \quad (1)$$

with $\omega=2\pi f$ the angular frequency, high frequency limit of permittivity ε_∞, relaxation strength $\Delta\varepsilon_k$, relaxation time τ_k and stretching exponents $0 \leq \beta_k,\ b \leq 1$ of the kth process and apparent direct current electrical conductivity σ_{DC}. Here, the highest frequency process will be denoted as IF (as intermediate process) whereas the lowest frequency will be denoted as LF (low frequency process)

The parameters obtained during the calibration were used as input in the scatter function model. The permittivity of the material in the probe will be now computed according to GDR model. In this framework, the 8 unknown parameters of the models were computed by fitting the scatter function. In Fig. 2, the comparison between the measured scattering function $S_{11}(\omega)$ and the best fit obtained with GDR is presented for sample S4. A good agreement can be observed with a RMSE equal to 0.0218. Same observations can be done for other sample with a RMSE equal to 0.0203 for S1, 0.0223 for S2, 0.0164 for S3, 0.0263 for S5 and 0.0292 for S6.

Finally, the relative complex permittivity can be computed from the obtained parameters with (1). The results are presented on Fig. 3 in terms of real part.

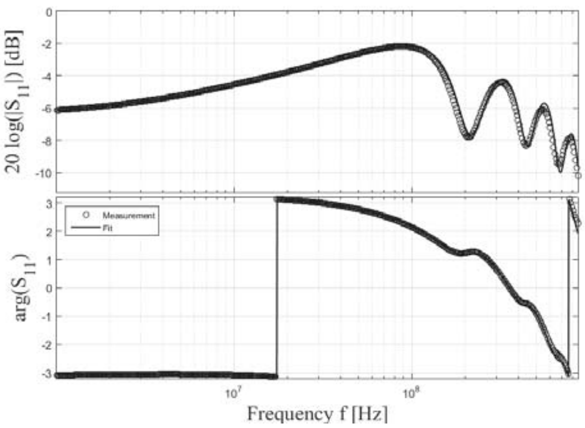

Fig. 2. Comparison between measured scatering function and best fit obtained with GDR model for sample S4.

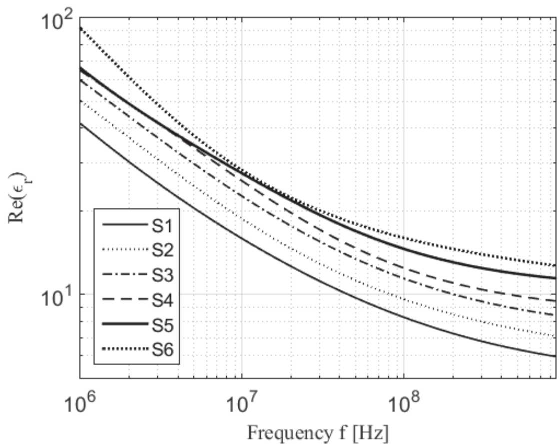

Fig. 3. Real part of estimated relative complex permittivity with GDR for the 6 samples.

The spectrums present a typical behavior of porous medium. We can observe a slight dispersion for the real part with values roughly between 100 and 6 over the frequency range. As expected, higher permittivity values are observed for higher water contents (see Table 1).

IV. Conclusions and Outlook

The presented method gives the opportunity to determine the frequency dependent relative complex permittivity of coarse grained materials up to 28 mm. The obtained spectrums represent fundamental data for high frequency electromagnetic measurement technique such as GPR. In the next step, other models for the frequency dependency of the complex permittivity such as mixing equations will be tested. The objective will be to achieve a simultaneous estimation of water content and density.

References

[1] M. Loewer, J. Igel, and N. Wagner, “Spectral Decomposition of Soil Electrical and Dielectric Losses and Prediction of in Situ GPR Performance,” IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens., vol. 9, no. 1, pp. 212–220, 2016.

[2] M. N. Soutsos, J. H. Bungey, S. G. Millard, M. R. Shaw, and A. Patterson, “Dielectric properties of concrete and their influence on radar testing,” NDT E Int., vol. 34, no. 6, pp. 419–425, 2001.

[3] A. Robert, “Dielectric permittivity of concrete between 50 Mhz and 1 GHz and GPR measurements for building materials evaluation,” *J.* Appl. Geophys., vol. 40, no. 1–3, pp. 89–94, 1998.

[4] D. A. Robinson, S. B. Jones, J. M. Blonquist Jr., and S. P. Friedman, “A Physically Derived Water Content/Permittivity Calibration Model for Coarse-Textured, Layered Soils,” Soil Sci Soc Am J, vol. 69, no. 5, pp. 1372–1378, 2005.

[5] H. Bhuyan, A. Scheuermann, D. Bodin, and R. Becker, “Use of Time Domain Reflectometry to Estimate Moisture and Density of Unbound Road Materials,” Transp. Res. Rec. J. Transp. Res. Board, vol. 2655, pp. 71–81, Jan. 2017.

[6] C.-P. Lin, “Analysis of nonuniform and dispersive time domain reflectometry measurement systems with application to the dielectric spectroscopy of soils,” Water Resour. Res., vol. 39, no. 1, 2003.

[7] J. a. Huisman, “Accuracy of frequency domain analysis scenarios for the determination of complex dielectric permittivity,” Water Resour. Res., vol. 40, no. 2, pp. 1–12, 2004.

[8] C. P. Lin, S. H. Tang, C. H. Lin, and C. C. Chung, “An improved modeling of TDR signal propagation for measuring complex dielectric permittivity,” J. Earth Sci., vol. 26, no. 6, pp. 827–834, 2015.

[9] U. Kaatze, “Reference liquids for the calibration of dielectric sensors and measurement instruments,” Meas. Sci. Technol., vol. 18, no. 18, pp. 967–976, 2007.

[10] T. Bore, M. H. Bhuyan, T. Bittner, V. Murgan, N. Wagner, and A. Scheuermann, “A large coaxial reflection cell for broadband dielectric characterization of coarse grained materials,” Meas. Sci. Technol., 2017.